

Problem Set 4

Applied Stats II

Due: April 10, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in .pdf form.
- This problem set is due before 23:59 on Friday April 10, 2026. No late assignments will be accepted.

Question 1

We're interested in modeling the historical causes of child mortality. We have data from 26855 children born in Skellefteå, Sweden from 1850 to 1884. Using the "child" dataset in the `eha` library, fit a Cox Proportional Hazard model using mother's age and infant's gender as covariates. Present and interpret the output.

My Answer:

I first loaded the data and created a survival object using the start time (birth), end time (death), and whether the event of death occurred within the timeframe of the study. Following these preprocessing steps, I fit the Cox Proportional Hazard model with the covariates `m.age` (mother's age) and `sex` (gender of child).

```
1
2 # load data on child mortality by mother's background and child gender
3 data("child", package = "eha")
4
5 # set survival of children
6 # start time, end time, whether the event happened or not
7 child_surv <- with(child, Surv(enter, exit, event))
8
9 # fit cox proportional hazard model
```

```

10 cox <- coxph(child_surv ~ m.age + sex, data = child)
11 summary(cox)

```

Table 1: Cox Proportional Hazard Model

	<i>Dependent variable:</i>
	Child Survival
Mother's Age	0.008*** (0.002)
Infant's Gender (1 = Female)	-0.082*** (0.027)
Observations	26,574
R ²	0.001
Max. Possible R ²	0.986
Log Likelihood	-56,503.480
Wald Test	22.520*** (df = 2)
LR Test	22.518*** (df = 2)
Score (Logrank) Test	22.530*** (df = 2)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Following implementation of the model, it can be said that on average, for every 1 unit increase in the mother's age there is a 0.008 increase in the expected log of the hazard for babies, holding infant's gender constant. In the context of gender of the baby, on average, there is a 0.082 decrease in the expected log of the hazard for female babies compared to male babies, holding the age of the mother constant.

These results indicate that as a mother's age increased, the likelihood of a child's death occurring throughout the time of the study was more likely, as well as that female babies had a higher likelihood of surviving throughout the study.

Question 2

We want to estimate how the amount of damage caused by a disaster relates to (1) whether disaster relief is provided and (2) the amount of relief that is provided conditional on a donation occurring when a disaster hits a given country. Estimate a Heckman selection model using the `disasters_response` data on GitHub in which `binContribution` is the outcome for the selection equation, and `originalContributionMillionUSDLogged` is the outcome for the second equation. Use `occurrences + deathsEM + normalizedDamageEMLogged` as your input variables for both equations. Present and interpret the output.

My Answer:

I began by loading the data and ran the Heckman selection model using `binContribution` as the outcome for the selection equation, measuring whether disaster relief is provided and `originalContributionMillionUSDLogged` as the outcome for the second equation, measuring the amount of relief provided conditional on a donation occurring when a disaster hits a given country.

By using the second equation, the way that an original donation given to the country following a disaster is considered in whether or not they receive subsequent disaster relief. The values from the second equation will help to indicate whether or not selection bias was present in the data generating process and OLS model.

```
1 # load data
2 disaster_data <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsII_
   2026/refs/heads/main/datasets/disaster_response.csv")
3
4 # heckman selection model
5 heck <- hekit(
6   selection = binContribution ~ occurrences + deathsEM +
7     normalizedDamageEMLogged,
8   outcome = originalContributionMillionUSDLogged ~ occurrences + deathsEM +
9     normalizedDamageEMLogged,
10  data = disaster_data
11 )
12 summary(heck)
```

From the output Table 2, it can be seen that the `invMillsRatio` value is not statistically significant. This value is also known as the *lambda* value and is used to determine if selection bias was present in the data generating process, and subsequently within the model. If *lambda* is statistically significant, there is selection bias present, but as it is not in this model, it does not appear that selection bias was present. This means that the selection model, which is also the original OLS model is sufficient to model this data. It can also be seen that none of the covariates from the outcome equation are statistically significant, yet they are all statistically significant in the selection model. This indicates that there in the outcome model there is not significant variation in the distribution

	Coefficients
S: (Intercept)	−1.30*** (0.02)
S: occurrences	0.02*** (0.00)
S: deathsEM	0.01*** (0.00)
S: normalizedDamageEMLogged	0.02*** (0.00)
O: (Intercept)	9.77 (6.04)
O: occurrences	−0.09 (0.05)
O: deathsEM	−0.03 (0.03)
O: normalizedDamageEMLogged	−0.11 (0.06)
invMillsRatio	−6.31 (3.42)
sigma	5.99
rho	−1.05
R ²	0.05
Adj. R ²	0.05
Num. obs.	49096
Censored	45848
Observed	3248

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 2: Heckman Selection Model